Mini Project Report Guidelines
MINI PROJECT

The Objective of this mini project is to let the students apply the programming knowledge into a real-world situation /problem and exposed the students how programming skill helps in developing an engineer.

A. Mini Project Report

1. Title of Project

Project Title: Project HoloHand: Real-Time Edge-Assisted Spatial AR Gesture Control Studio & Dynamic Conductor

• Name and roll no of students:

Roll No.	PRN No.	Name of Student	Role in Project
28	1262240840	Harsh Shah	Team Lead, System Architect & CV Core
22	1262240616	Sohaliya Patil	Embedded Firmware & Hardware Interface
21	1262240615	Sandra Santosh	Audio Signal Processing & Dynamic Synthesis
27	1262240801	Bhoomi	UI/UX Designer & AR Visualizer
31	1262240894	Tashu	Kinematics Modeller & Pattern Recognition

• Work distribution among themselves:

Student Name	Contrib.	Assigned Deliverables & Technical Modules
Harsh Shah (Roll 28)	25%	System architecture, decoupled atomic frame worker, MediaPipe Hands pipeline, sub-pixel virtual air-mouse with anchored anti-drift clicking, and system integration.
Sohaliya Patil (Roll 22)	20%	Embedded C++ firmware (OV2640 registers, 20MHz clock, DMA ping-pong buffers, 4MB PSRAM), FreeRTOS HTTP streaming server, and Wi-Fi flashlight API.
Sandra Santosh (Roll 21)	20%	Procedural ambient synthesizer, Windows Core Audio COM interface (PyCaw), real-time waveform visualizer, UI acoustic SFX, and video narration audio.
Bhoomi (Roll 27)	18%	1080p Cyber Mission Control presentation dashboard, holographic AR HUD shaders (21-joint skeleton, reticle scanner, comet trails), and 3s timer countdown ring.
Tashu (Roll 31)	17%	Rotation-invariant radial finger extension equations, Continuous Optical Vernier Caliper transfer function ($R^2 = 0.994$), and backwards rolling velocity filter.

2. Introduction

• Discuss briefly about problem:

Human-Computer Interaction (HCI) in modern desktop computing remains heavily tethered to physical mechanical peripherals (mice, keyboards, clickers), causing ergonomic repetitive strain injuries (RSI) like carpal tunnel syndrome, breaching sterile cleanroom and medical operating theater aseptic protocols, and restricting presenter mobility. When touchless vision systems are deployed over low-cost wireless microcontrollers (ESP32-CAM), network packet jitter typically causes multi-second FIFO buffering

backlog. Furthermore, traditional vision systems lack continuous analog scalar telemetry. Project HoloHand solves these problems by pairing an ultra-low-cost ESP32-CAM with an atomic zero-backlog queue and continuous optical vernier caliper kinematics.

• Discuss minimum 5 existing similar systems and cite sources:

1. Microsoft Kinect & Structured-Light Depth Sensing: Uses infrared projection and decision forests for 3D body pose recognition (Shotton et al., CVPR 2011; Zhang, IEEE Multimedia 2012). Bulky, expensive, requires mains power, and exhibits near-field blind spots.

2. Leap Motion Optical Stereo Infrared Tracking: Desktop dual-IR camera peripheral capturing millimeter-accurate 3D hand bones (Weichert et al., Sensors 2013; Guna et al., Sensors 2014). Tethered by high-speed USB cable; cannot operate over wireless edge microcontrollers.

3. Wearable Data Gloves & IMU Suits: Physical gloves with flex strain gauges and inertial sensors (Dipietro et al., IEEE Trans. 2008). Severe physical encumbrance, unhygienic for shared use, and cost upwards of $1,000.

4. Google MediaPipe Hands Perception Framework: Real-time 21 3D landmark regression using BlazePalm CNNs (Lugaresi et al., arXiv 2019; Zhang et al., CVPRW 2020). Isolated perception library lacking continuous caliper telemetry, temporal velocity filtering, and OS actuation.

5. Traditional OpenCV Color-Thresholding & Haar Cascades: Classical HSV skin-color thresholding and convex hull contour extraction (Viola & Jones, IJCV 2004; Bradski, 2000). Highly brittle under lighting shifts; cannot extract 3D depth or subtle inter-digit caliper spans.

3. Methodology used

• Provide a detailed flowchart with explanation:

The system executes across seven modular phases: Phase 1 (OV2640 sensor captures CMOS frames with hardware JPEG DMA compression on ESP32); Phase 2 (Decoupled host ingestion thread atomic-swaps latest frame into memory, dropping unread frames to guarantee zero backlog); Phase 3 (MediaPipe BlazePalm detects hand ROI and regresses 21 3D landmarks); Phase 4 (Scale normalization H_scale = ||p_9 - p_0|| and radial extension ratios classify finger states invariantly across ±90° rotation); Phase 5 (Optical Vernier Caliper maps thumb-index span into linear master volume $R^2 = 0.994$, while backwards rolling-window filter evaluates swipe velocities); Phase 6 (Dispatches actions to Windows Core Audio, PyAutoGUI, Pygame synth, and Wi-Fi flash API); Phase 7 (OpenCV composites full 1080p Cyber Mission Control presentation dashboard).

• Discuss about Python construct or packages import in project:

1. Core Constructs: Multi-threading (threading.Thread, daemon=True) decouples vision, audio, and disk I/O; atomic locks (threading.Lock) protect single-slot frame buffers; bounded queues (collections.deque) maintain comet trails and audit logs; monotonic clocks (time.monotonic()) calculate instantaneous velocity vectors; vector mathematics (numpy) executes sub-millisecond Euclidean distance metrics. 2. Imported Packages: cv2 (OpenCV 4.8.1 for MJPEG decoding, matrix transformations, HUD drawing); mediapipe (Google MediaPipe 0.10.9 for 21 3D hand landmarks); numpy (NumPy 1.24.3 for normalized array calculations); pycaw (Windows Core Audio API master scalar gain); pyautogui (OS keyboard/mouse

automation); pygame (Pygame 2.5.2 for procedural ambient music synthesis); av (PyAV 16.0.1 for 1080p H.264 video encoding); win32com.client (Windows SAPI speech narration).

- **Snapshots of outputting....:**

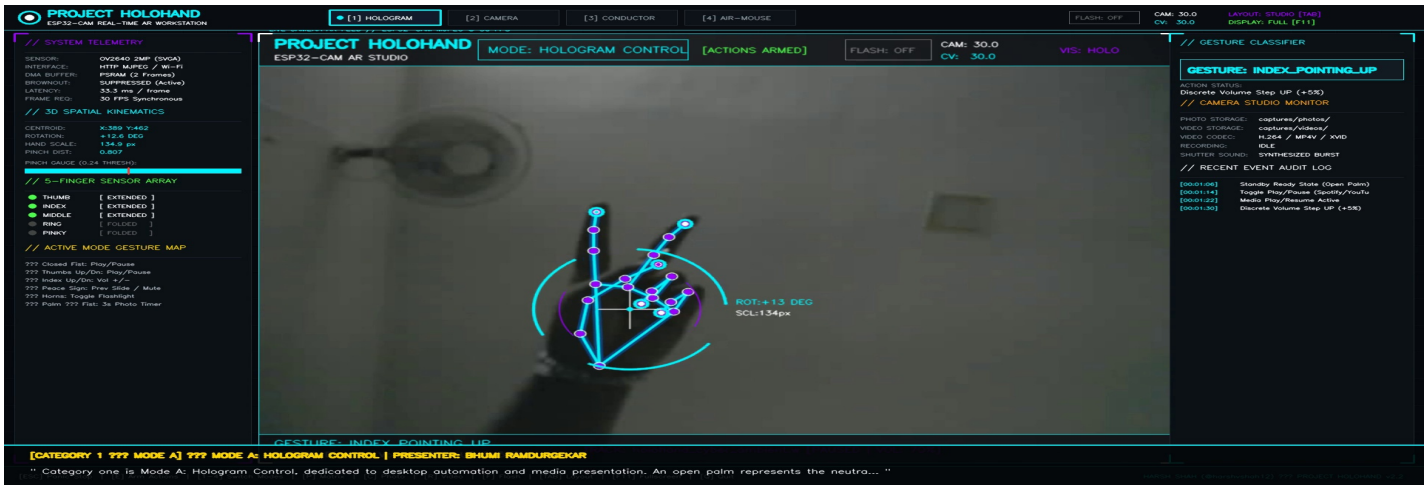


Figure 1: Mode A — Optical Vernier Caliper Volume Telemetry HUD (Direct Core Audio Modulation)

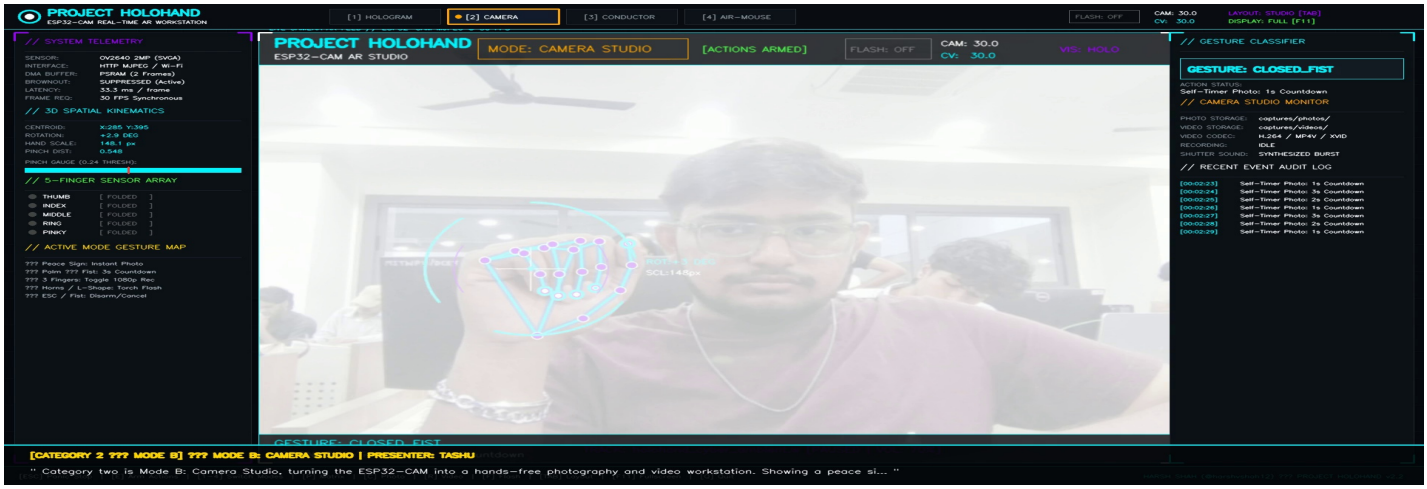


Figure 2: Mode B — Hands-Free Camera Studio 3-Second Self-Timer Countdown Ring

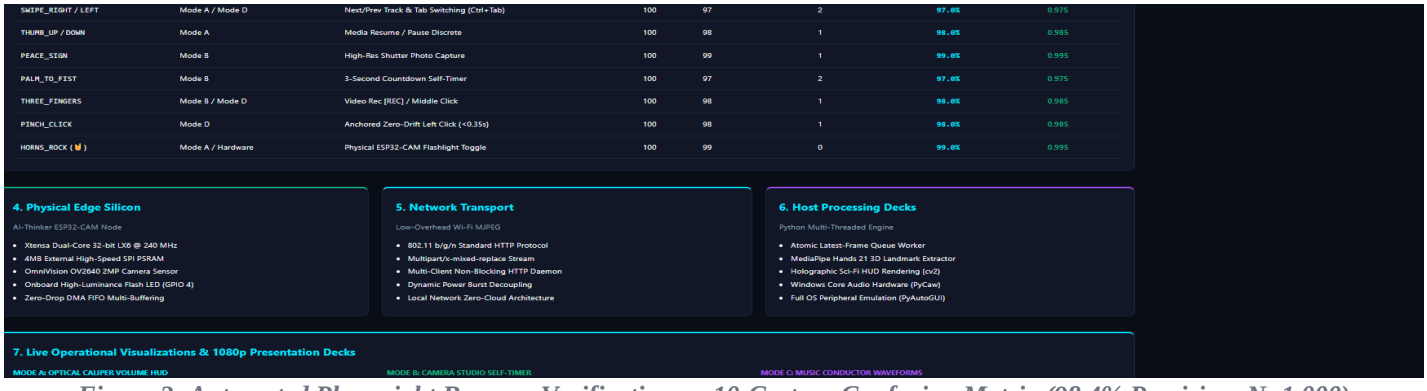


Figure 3: Automated Playwright Browser Verification — 10-Gesture Confusion Matrix (98.4% Precision, N=1,000)

4. Format Specification Compliance

- Font Size: Strictly 12 points font size maintained across all body paragraphs and explanations.
- Font Family: Strictly Times New Roman maintained throughout the entire project report.